
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Reinforcement learning (RL) has become a central driver of stronger reasoning in
2 language models, yet its training dynamics remain insufficiently explained. We
3 provide a concrete empirical and theoretical account of how RL changes reasoning
4 behavior in LLMs. Through systematic reasoning-pattern-level and token-level
5 analysis across training, we establish that individual reasoning patterns maintain
6 relatively stable success rates, while RL concentrates its updates on a sparse set of
7 critical tokens and thereby reshapes the distribution over reasoning patterns that the
8 model selects. Building on this finding, we develop a theoretical framework that
9 characterizes RL dynamics under two representative reward mechanisms: verifiable
10 reward (RLVR) and the model’s internal feedback (RLIF). For RLVR, our analysis
11 identifies two distinct regimes: models with sufficient initial reasoning quality
12 converge readily toward optimal reasoning strategies, whereas weaker models
13 encounter optimization difficulty in entangled states. For RLIF, we show why
14 internal rewards can produce early performance gains and why continued training
15 can also drive degradation below the base model. Extensive experiments support
16 these conclusions, positioning reasoning-pattern selection as a decisive mechanism
17 behind RL-based language-model enhancement.

18 1 Introduction

19 Recent state-of-the-art reasoning models, including Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1
20 [14], deliver exceptional performance on complex logical tasks such as mathematics [30, 44, 41] and
21 programming [49, 40, 8]. Reinforcement learning (RL) is a core post-training mechanism behind
22 these gains and has become a decisive tool for elevating model reasoning capabilities.

23 This success has made the mechanism of RL post-training a central question for LLM research. By
24 comparing pass@k performance, Yue et al. [42] show that RL-post-trained models struggle to surpass
25 base models, indicating that RL does not necessarily create fundamentally new reasoning patterns.
26 From an entropy perspective, Cui et al. [9], Zhang et al. [45] prove that RL-based methods can reduce
27 policy entropy, and Cui et al. [9] further connects policy entropy to model performance empirically.
28 Wang et al. [33], Huan et al. [17] demonstrate that RL concentrates optimization on a sparse subset
29 of critical tokens. These results identify important pieces of the mechanism, but they still leave the
30 training dynamics of RL only partially characterized at both empirical and theoretical levels.

31 We address this gap by conducting a systematic reasoning-pattern-level and token-level analysis
32 throughout RL training. Rather than only measuring aggregate performance, we track next-token
33 ranking shifts and use both LLM-based and rule-based procedures to extract, classify, and measure
34 reasoning patterns in model responses. This analysis provides direct evidence that RL primarily
35 optimizes a sparse subset of critical tokens and thereby *reshapes reasoning pattern distributions* to
36 change model performance. At the same time, the intrinsic success rates of individual patterns remain
37 relatively stable during training. Compared with prior analyses [17, 42], these results give a sharper

38 empirical account of what RL changes: it selects among existing reasoning patterns more effectively
39 rather than uniformly improving every pattern.

40 Guided by these findings, we develop a theoretical framework that represents reasoning as a two-stage
41 question-reason-answer process $q \rightarrow r \rightarrow a$: the model first selects a reasoning pattern according
42 to $\pi(r|q)$ and then generates an answer according to $\pi(a|r, q)$. This abstraction directly exposes the
43 training dynamics of two representative reward types: verifiable reward (RLVR) [14, 30] and the
44 model’s internal feedback (RLIF) [47, 4]. For RLVR, we prove convergence toward the reasoning
45 pattern r^* with the highest success rate under our setting, matching the empirical pattern-selection
46 behavior. We further separate two convergence regimes: strong base models rapidly move toward
47 optimal patterns, whereas weaker base models face optimization difficulty in entanglement stages.
48 For RLIF, we explain why internal rewards can improve performance early in training and why the
49 same mechanism can eventually converge to a state with worse performance than the base model.
50 Additional experiments reinforce the theoretical account.

51 This paper makes the following contributions:

- 52 • We deliver a systematic reasoning-pattern-level and token-level account of RL training. By
53 combining next-token prediction ranking shifts with LLM-based and rule-based reasoning-pattern
54 analysis, we show more directly than prior work [17, 42] that RL concentrates optimization on a
55 sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* while leaving the
56 intrinsic success rates of individual patterns relatively stable.
- 57 • We introduce a formal two-stage mathematical framework that models reasoning as $q \rightarrow r \rightarrow a$
58 (question-reason-answer) and use it to characterize training dynamics for two representative RL
59 approaches. For RLVR, we prove convergence to the reasoning pattern with the highest success rate
60 and identify the contrast between rapid convergence for strong base models and entanglement-stage
61 optimization challenges for weaker models. For RLIF, we justify early-stage performance gains
62 and explain why continued optimization can ultimately underperform the base model.
- 63 • We substantiate the theory with additional case studies and numerical simulations. The resulting
64 framework bridges empirical observations and formal training-dynamic analysis, and provides
65 actionable insight for designing more reliable RL-based LLM post-training.

66 2 Related Works

67 **Reinforcement Learning for LLMs** Reinforcement learning has become a leading post-training
68 approach for strengthening large language models (LLMs), especially for aligning models with human
69 preferences [25, 48] and improving performance on demanding mathematical and programming tasks
70 [30, 20, 22]. The reward signal is central to this success. Traditional RL pipelines often rely on
71 extensively trained reward models built from human-annotated data [26, 25, 2], whereas recent work
72 increasingly exploits rewards that are cheaper, more direct, and more verifiable. Representative exam-
73 ples include reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement
74 learning with internal feedback (RLIF) [47, 4]. This shift has also driven the design of specialized
75 RL algorithms, including GRPO [30], DAPO [41], and VAPO [43].

76 **Theoretical Analyses and Mechanism Interpretation of RL for LLMs.** The empirical success of
77 RL in LLMs has produced a growing line of theoretical and mechanistic studies. Early work largely
78 emphasized high-level questions such as reward design [28, 18, 38] and new training algorithms
79 [37, 11, 21]. As the field has moved toward simpler reward structures such as RLVR and RLIF, the key
80 question has shifted from whether RL works to why these reward signals reliably change reasoning
81 behavior. Shao et al. [29] show that spurious rewards can improve reasoning by unlocking latent skills
82 from pretraining, while Agarwal et al. [4] explain performance gains through entropy minimization.
83 Gandhi et al. [13] identify cognitive behaviors, including backtracking, that correlate with stronger
84 reasoning. Complementary evidence from Zhao et al. [46] shows that RL fine-tuning amplifies
85 pre-trained behaviors and drives convergence toward dominant output formats, a phenomenon further
86 studied by Wang et al. [34], Cui et al. [10] through entropy collapse. Our work sharpens this
87 mechanistic picture by making reasoning-pattern selection the central object of analysis and by
88 deriving a mathematical training-dynamics account that explains the observed pattern shifts.

89 3 Preliminaries

90 **Reinforcement Learning for LLMs** We view the language model π_θ , parameterized by θ , as the
 91 policy optimized during post-training. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy
 92 generates an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$. The RL objective takes the following standard form:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \| \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

93 where π_{ref} denotes the base reference policy and β controls the KL penalty that keeps π_θ from
 94 drifting excessively away from π_{ref} .

95 The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ can be instantiated in several ways, including a trained reward model
 96 [25] or a rule-based scoring function [30]. We focus on two representative and increasingly important
 97 settings: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4].

98 In RLVR, $r_\phi(\mathbf{x}, \mathbf{y})$ directly scores whether the answer $\mathbf{y}$ matches the correct answer to question $\mathbf{x}$. A
 99 canonical binary reward is:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} \text{ is the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

100 RLIF [47, 4] instead uses rewards derived solely from the model’s own predictive behavior. Following
 101 Zhao et al. [47], we instantiate the RLIF reward as the negative average KL divergence between a
 102 uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s next-token distribution:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \| \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

103 Our analysis targets the training dynamics induced by both RLVR and RLIF, using empirical evidence
 104 and theoretical characterization as complementary tools.

105 4 Experimental Exploration for RL Training

106 We analyze RL training through a sequence of experiments that isolate how different rewards change
 107 model behavior. The training overview shows a sharp contrast: RLVR produces steady performance
 108 gains, whereas RLIF can become unstable and even degrade performance. The reasoning-pattern
 109 analysis then identifies the source of this gap: RLVR increasingly selects reasoning patterns with
 110 higher success rates, while RLIF fails to consistently specialize in those patterns. Finally, token-level
 111 analysis reveals the operational mechanism behind these pattern shifts, showing that RL changes
 112 the probability ranks of only a surprisingly small fraction of tokens. Compared with previous work
 113 [17, 42], these experiments provide a more direct account of RL training dynamics and demonstrate
 114 that RL primarily optimizes a sparse subset of critical tokens, thereby *reshaping reasoning pattern*
 115 *distributions* to affect model performance.

116 4.1 Training Procedure Overview for RLVR and RLIF

117 We first examine the full training trajectories of RLVR and RLIF. To make the comparison controlled,
 118 we use Qwen2.5-3B [39] as the base model, train on the MATH dataset [16], and keep all other
 119 settings identical across the two algorithms.

120 Figure 1a shows that the two reward mechanisms induce clearly different trajectories. RLVR yields a
 121 stable process in which model performance improves continuously and gradually converges. RLIF, in
 122 contrast, can deliver early gains but then loses performance under continued training, in some runs
 123 falling below the original base model. This divergence makes fine-grained analysis necessary: the
 124 reward type does not merely affect optimization speed, but changes the direction in which reasoning
 125 behavior evolves.

126 4.2 Reasoning-pattern Level Analysis

127 To explain the performance gap between RLVR and RLIF, we analyze training at the reasoning-pattern
 128 level. We first collect responses from the base model and use GPT-4o [19] to group them into distinct

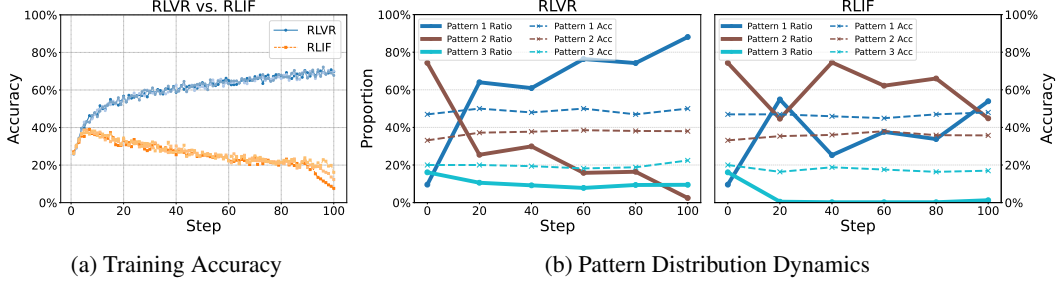


Figure 1: **(a)** Training procedures for RLVR and RLIF on the MATH dataset. RLVR produces stable gains, whereas RLIF first improves performance and then can drive a decline. We run multiple trials for each training paradigm by varying the random seed. **(b)** Reasoning-pattern level comparison. The solid line gives each pattern’s response proportion among all patterns (left vertical axis), and the dotted line gives the corresponding accuracy (right vertical axis). RLVR progressively shifts mass toward higher-accuracy patterns and away from lower-accuracy patterns, while RLIF exhibits unstable pattern dynamics. For readability, we sort and name patterns from high to low according to their corresponding accuracy.

categories according to keywords and logical structure, producing a description for each pattern. We then apply these classifications across training to measure both the distribution of reasoning patterns and the success rate associated with each pattern.

We specifically track how pattern distributions and pattern accuracies evolve during both RLVR and RLIF training. Figure 1b reports the Number Theory tasks from the MATH dataset [16], with additional models and datasets reported in Appendix A. The analysis establishes three findings:

- RLVR-trained models consistently shift toward reasoning patterns with higher success rates, directly explaining the steady improvement in overall accuracy.
- RLIF produces unstable reasoning-pattern distributions and does not reliably specialize in the more effective patterns.
- The success rate of each individual reasoning pattern remains stable throughout training for both methods.

These results identify reasoning-pattern dynamics as the key source of the performance difference between RLVR and RLIF. We therefore turn to token-level experiments to determine how RL produces these distributional shifts.

4.3 Token-level Analysis

To uncover the mechanism behind the observed reasoning-pattern dynamics, we conduct a token-level analysis inspired by Huan et al. [17], Yue et al. [42]. We sample base-model responses for questions from GSM8K [6], MATH [16], and AIME [7]. For these identical responses, we compare token ranks under the base model and the RL-enhanced model, where a token’s rank is the position of its probability among all vocabulary tokens given the preceding context. Table 1 shows that individual token ranks remain largely stable: in most settings, fewer than 10% of tokens experience a rank shift. Thus, RL does not broadly rewrite the entire reasoning trajectory; it selectively changes the probabilities of a sparse set of critical tokens that steer pattern selection.

Together, the experiments establish a clear mechanism for RL training dynamics: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains relatively stable during training.*

Table 1: Token ranking-change ratio after RL training.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

These empirical findings motivate the mathematical framework in the next section: RL performance changes can be analyzed through the interaction between sparse token updates and the resulting selection over reasoning patterns.

5 Theoretical Characterization of RL Training Dynamics

The empirical results identify reasoning-pattern selection as the primary behavioral change induced by RL. We now turn this observation into a theoretical characterization of the training process. Our framework abstracts model reasoning into two steps: selecting a reasoning pattern and then deducing the final answer, with RL primarily acting on the selection step. This abstraction lets us analyze how different rewards shape training dynamics. For RLVR, we characterize two regimes with distinct optimization behavior. For RLIF, we explain both its initial performance gains and its potential to converge to models that underperform the base model.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We formalize reasoning as a two-stage stochastic process. Given a question q , the model first selects a reasoning pattern r from candidate patterns $\mathcal{R} = \{r_1, r_2, \dots\}$ and then generates a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ conditioned on that pattern. The model selects r_i with probability $p(r_i|q)$, and each pattern has its own success rate. Under this abstraction, the RLIF objective (Eq. 3.3) becomes¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

Policy Parameterization. Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ denote the set of output sequences. Given a question q , the language model defines a distribution over output sequences $y \in \mathcal{Y}$ autoregressively:

$$\text{(general policy)} \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is parameterized by θ , and the model predicts the l -th token from the previous context $y_{<l}$. The next token follows $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ for $z = f_\theta(x, y_{<l})$.

Because practical LLMs are highly complex, direct optimization guarantees for RL training dynamics are difficult to obtain [3, 23]. Following prior theoretical analyses [27, 24, 10, 45], we therefore study a *tabular policy* parameterization. This policy is a special case of Eq. 5.2 in which each output receives a trainable logit for the corresponding previous token, i.e., for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$\text{(tabular policy)} \quad \pi_\theta(y_l|x, y_{<l}) = \pi_\theta(y_l|y_{l-1}) = \text{softmax}(\theta_{:,y_{l-1}})_{y_l}, \quad (5.3)$$

where $\theta_{:,y_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ associated with y_{l-1} .

Optimization for RL. To analyze training dynamics, we combine the tabular policy in Eq. 5.3 with the RL objective $\phi_{\text{RL}}(\theta)$ in Eq. 3.1. We study the small-learning-rate limit through gradient flow:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

where $\theta(t)$ denotes the policy parameters at training time t , initialized at $\theta(0) = \theta_{\text{ref}}$.

Our experiments show that individual reasoning-pattern success rates remain stable during training. This motivates a structural simplification: RL primarily changes the mapping from questions to reasoning patterns, while the conditional mapping from a selected pattern to the final answer changes much less. We formalize this empirically supported behavior as follows:

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

197 **Assumption 5.1.** *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer*
 198 *$\mathbf{r}^*$, defined as $\pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

199 For the rest of the section, we work under Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$ for the
 200 fixed success rate of pattern $\mathbf{r}$. We first derive the optimal policy for the RL objective (Eq.3.1) under
 201 the general autoregressive policy (Eq.5.2), and then analyze RLVR and RLIF training dynamics under
 202 the tabular policy (Eq. 5.3).

203 5.2 The Optimal Policy for RL

204 Our empirical results show that RLVR steadily improves performance by incentivizing higher-success
 205 reasoning patterns, whereas RLIF produces less stable improvements. We first explain this contrast
 206 through the optimal policies induced by the RLVR and RLIF objectives:

207 **Proposition 5.2.** *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive*
 208 *policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

209 where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the
 210 reasoning path reward. Specifically:

- 211 • For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- 212 • for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution,
 213 i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

214 Proposition 5.2 separates the optimization targets of RLVR and RLIF. Assuming π_{opt} lies in the
 215 general autoregressive policy class, which holds for LLMs and for the tabular policies analyzed below,
 216 RLVR increases the probability of a reasoning pattern $\mathbf{r}$ whenever its probability of deducing the
 217 correct answer $p^*(\mathbf{r})$ is sufficiently high, i.e., $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$. RLIF is fundamentally different:
 218 $R_{\text{RLIF}}(\mathbf{r})$ measures confidence in the final-answer distribution but does not distinguish the correct
 219 answer $\mathbf{a}^*$ from incorrect alternatives. Therefore, the RLVR optimum is aligned with better accuracy,
 220 while the RLIF optimum is not guaranteed to improve over the base model.

221 Moreover, since β is typically small (e.g., 0.001) in practice [26, 5, 44], as $\beta \rightarrow 0$, we have:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

222 In this limit, the policy becomes deterministic and always selects the reasoning pattern $\mathbf{r}$ with
 223 the largest $R(\mathbf{r})$, regardless of the initial reference-model distribution. The remaining question is
 224 dynamic rather than static: how does RLVR reach high-reward patterns, and why can RLIF improve
 225 early despite its misaligned optimum? We answer these questions by analyzing both methods under
 226 the tabular policy in Eq. 5.3.

227 5.3 Training Dynamic Analysis for RLVR

228 Our experiments show that RLVR gains depend strongly on the base model’s initial reasoning quality.
 229 We therefore focus on the probability of the *optimal reasoning pattern*, namely the pattern $\mathbf{r}^*$ with
 230 the highest success rate for reaching the correct answer $\mathbf{a}^*$, $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$. Gradient-flow
 231 analysis reveals two regimes. In the first, $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ steadily increases and converges to 1. In the
 232 second, training initially enters an entanglement stage in which a suboptimal pattern $\mathbf{r}' \neq \mathbf{r}^*$ slows
 233 optimization toward the optimal pattern. Once the model exits this stage, the dynamics transition to
 234 the first regime and converge to $\mathbf{r}^*$.

235 **Theorem 5.3** (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq*
 236 *3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1*
 237 *holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$,*
 238 *defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$, satisfies:*

$$\text{(Regime 1)} \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

239 then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

240 Theorem 5.3 covers the favorable case in which the base reference model is already strong enough
 241 that its overall accuracy exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7). In
 242 this regime, RLVR efficiently drives the model toward the optimal reasoning pattern at rate $\mathcal{O}(1/\epsilon)$,
 243 yielding overall accuracy close to the success rate of that optimal pattern.

244 Practical RLVR training can nevertheless be difficult [44, 36], especially when the base reference
 245 model is weaker. The second regime captures this case: the model first enters an entanglement
 246 stage in which the *suboptimal reasoning pattern with the second-highest success rate*, defined as
 247 $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, slows the increase of the optimal pattern:

248 **Theorem 5.4** (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq*
 249 *3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1*
 250 *holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of*
 251 *the initialized model satisfies:*

$$(Regime\ 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

252 then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|q)} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|q)}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|q)} \quad (5.9)$$

253 with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can
 254 guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} >$
 255 $p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^* \text{ for } t \geq T_0$.

256 Theorem 5.4 analyzes a special regime in which only the optimal and second-best reasoning patterns
 257 have success rates above the model’s overall accuracy. We use this setting to keep the analysis
 258 tractable, and the same mechanism is expected to extend to more general cases where *at least* two
 259 reasoning patterns exceed the average, though the proof would become more involved.

260 The critical implication of Theorem 5.4 is that RLVR may need T_0 time steps before the overall
 261 success rate exceeds that of the suboptimal pattern and the process reaches the favorable regime of
 262 Theorem 5.3. Although Theorem 5.3 gives a convergence time T_1 that is polynomial in $1/\epsilon$, the
 263 transition time T_0 in Theorem 5.4 can grow super-exponentially with $\gamma_{\pi_{\text{ref}}}$ —the ratio between the
 264 total success rate through $\mathcal{R}/\{\mathbf{r}'\}$ and the success rate through $\mathbf{r}^*$. When the base model assigns very
 265 small probability to the optimal reasoning pattern, $\gamma_{\pi_{\text{ref}}}$ can be large, making T_0 prohibitively large.
 266 This result formalizes the entanglement stage: the model may require a very long training period
 267 before it selects the optimal pattern with substantial probability.

268 Sec 6 provides additional experiments and case studies that illustrate both regimes.

269 5.4 Theoretical Explanation for RLIF

270 Proposition 5.2 shows that RLIF optimizes confidence in the final-answer distribution without distin-
 271 guishing the correct answer $\mathbf{a}^*$ from incorrect alternatives. This misalignment explains why RLIF
 272 can eventually degrade performance relative to the base model, matching our empirical observa-
 273 tions. At the same time, prior studies show that RLIF can improve performance without external
 274 rewards Agarwal et al. [4], Zhao et al. [47]. We now explain why these early gains are possible for a
 275 well-trained LLM.

276 We start from a well-trained LLM satisfying the following assumption:

277 **Assumption 5.5.** *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all*
 278 *possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|q, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|q) = \mathbf{a}^*$.*

279 Because majority voting [35] is an effective way to improve model performance, well-trained LLMs
 280 typically assign higher aggregate probability to the correct answer $\mathbf{a}^*$ than to incorrect answers;
 281 Assumption 5.5 therefore captures a plausible property of modern LLMs. Under this condition, we
 282 study the case $|\mathcal{A}| = 2$, with uniformly distributed reasoning-path success rates and high-entropy
 283 reasoning-pattern selection². We then analyze the overall accuracy dynamics at $t = 0$ and obtain:

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

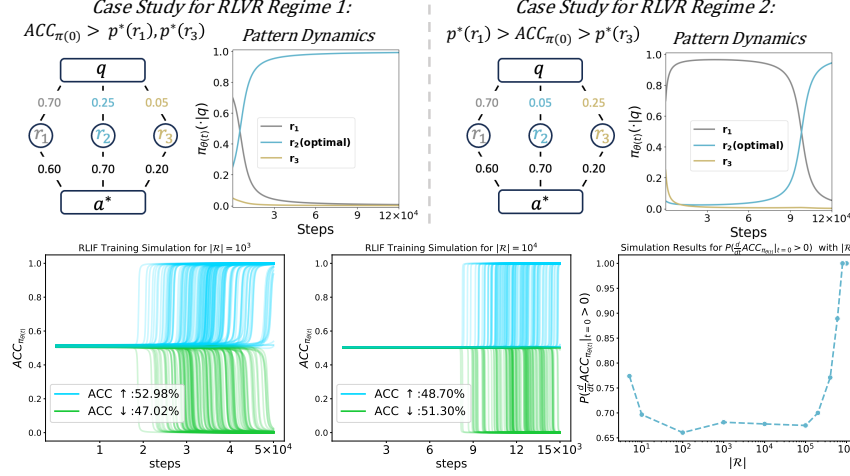


Figure 2: Case studies for RLVR (upper) and training simulations for RLIF (bottom). The upper panels exhibit the two RLVR regimes predicted by the theory: rapid convergence (left) and an entanglement phase (right). The bottom panels report, across varying $|\mathcal{R}|$, the probabilities that RLIF converges to different states (left) and the probability that it produces initial performance gains (right).

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.
2. With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.

The first result in Theorem 5.6 proves that overall accuracy increases at initialization, giving a direct theoretical explanation for RLIF’s early performance gains. The long-run behavior is different. Proposition 5.2 implies that when $\beta = 0$, the policy converges deterministically to the reasoning pattern $\mathbf{r}$ that maximizes $R_{\text{RLIF}}(\mathbf{r})$. The second result in Theorem 5.6 shows that, with probability 0.5, this maximizing pattern is also the pattern with the minimum success rate: $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$. Thus, although RLIF initially improves accuracy, continued training can push the model toward very low-accuracy reasoning paths and produce performance below the base model.

6 Case Studies and Numerical Simulations

We further substantiate the theory with case studies and training simulations that directly instantiate the main theorems. For RLVR, Fig. 2 (upper) displays the two regimes characterized in Theorems 5.3 and 5.4. For RLIF, Fig. 2 (bottom) reports simulations in which each reasoning path’s success rate follows a uniform distribution, validating the predicted behavior at both initialization and convergence.

6.1 Case Studies for RLVR

Section 5.3 identifies two regimes in RLVR training dynamics. In the first regime (Eq 5.7), the probability of selecting the optimal reasoning pattern $\pi_{\theta}(\mathbf{r}^*|q)$ increases monotonically until convergence. In the second regime (Eq 5.8), the model first passes through an entanglement phase and requires a substantially longer transition before entering the rapid-convergence dynamics of the first regime. We verify both regimes:

- Fig. 2 (upper left) confirms the rapid-convergence regime: when the reference model satisfies Eq 5.7, so that its overall accuracy exceeds the success rates of all non-optimal reasoning patterns, the model quickly concentrates on the optimal reasoning pattern.
- Fig. 2 (upper right) confirms the entanglement regime in Theorem 5.4. When a suboptimal reasoning pattern $\mathbf{r}'$ has higher success rate than the initial overall accuracy, the model first enters an entanglement phase. The transition period T_0 defined in Theorem 5.4 must pass before the rapid optimization dynamics of case 1 begin. The displayed scenario has large $\gamma_{\pi_{\text{ref}}}$ (defined in Eq 5.9),

315 with $\gamma_{\pi_{\text{ref}}} = 6$ because $(0.25 + 0.05)/0.05 = 6$. Since T_0 scales with $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$, this setting yields a
 316 prohibitively long transition and substantially delays convergence to the optimal reasoning pattern.
 317 The observed pattern dynamics therefore match the theoretical entanglement prediction.

318 6.2 Numerical Simulations For RLIF

319 Section 5.4 shows that, under Assumption 5.5 and the base-model distribution in Theorem 5.6, the
 320 accuracy derivative at $t = 0$ is positive with probability 1 as $|\mathcal{R}| \rightarrow \infty$, while the model has a 50%
 321 probability of converging to the least accurate reasoning path. The simulations confirm both claims.
 322 Fig. 2 (bottom left) shows RLIF simulations across varying $|\mathcal{R}|$: the probability of convergence to an
 323 improved state ($ACC \uparrow$) is approximately equal to the probability of convergence to a degraded state
 324 ($ACC \downarrow$), matching the second result in Theorem 5.6. Fig. 2 (bottom right) shows that, for large
 325 $|\mathcal{R}|$, the probability of initial performance improvement approaches 1, confirming the theorem’s first
 326 result.

327 7 Conclusions and Limitations

328 This work provides a decisive empirical and theoretical account of how reinforcement learning
 329 changes reasoning in LLMs: it primarily reshapes the distribution over reasoning patterns rather than
 330 uniformly improving every pattern. We develop a mathematical framework for two representative
 331 reward mechanisms, RLVR and RLIF, and validate its predictions through case studies and simulations.
 332 The results explain why RLVR can reliably concentrate probability on high-success reasoning patterns,
 333 why weak base models can face an entanglement stage, and why RLIF can improve early yet degrade
 334 with continued training. Several limitations remain important future directions: LLM-identified
 335 reasoning patterns require further interpretability validation, the theoretical framework should be
 336 extended to richer real-world reasoning settings, and the analysis should be generalized beyond the
 337 specific base-model assumptions studied here.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

Section 4.2 analyzes reasoning-pattern dynamics on Number Theory tasks from MATH. We extend that analysis to Geometry tasks in Figure 3. The same pattern emerges: RLVR progressively shifts toward higher-accuracy reasoning patterns, whereas RLIF produces unstable pattern selection. This additional evidence reinforces the empirical and theoretical conclusions in the main paper.

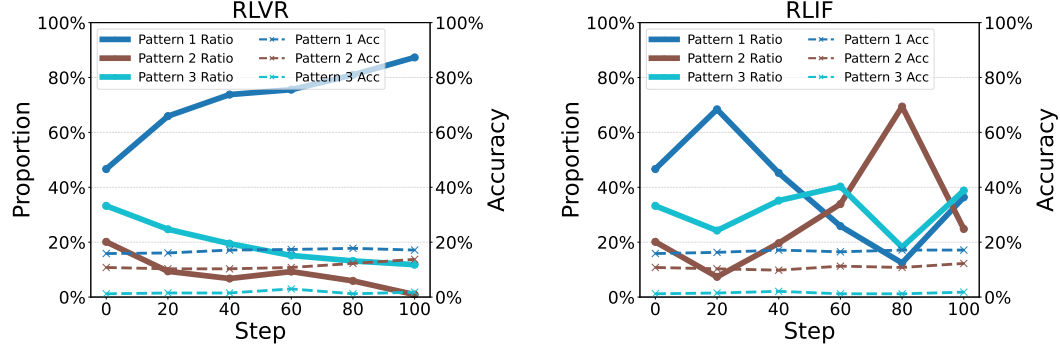


Figure 3: Dynamics of pattern distributions and corresponding accuracies for RL-enhanced models on Geometry tasks. RLVR concentrates on the highest-accuracy pattern (Pattern 1), whereas RLIF remains unstable.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To test whether the findings transfer across models, we conduct supplementary experiments summarized in Figure 4. We use Qwen-2.5-7B-Instruct [39] as the base model and compare it with Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source RLVR-enhanced variant for mathematical reasoning. The evaluation covers number theory, geometry, algebra, calculus, and counting and probability using challenging problems sampled from MATH [16], and it further includes complex mathematical problems from AMC23 [1]. The results consistently support our claim: RLVR increases the frequency of high-accuracy reasoning patterns and reduces the use of less effective patterns. Detailed task-specific analyses appear in Appendix B.3.

Across these tasks, reasoning-pattern analysis shows that the highest-success patterns consistently become more prevalent after RLVR enhancement, reinforcing the conclusions of Section 4 and Section 5. In tasks such as Algebra and AMC23-19 in Figure 4, individual pattern success rates remain consistent, further supporting Assumption 5.1 in the theoretical analysis.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To further assess the theory at larger scale, we compare QwQ-32B [32] with its base model, Qwen-2.5-32B-Instruct [39]. QwQ-32B substantially strengthens reasoning through RLVR. We evaluate reasoning-pattern shifts on four tasks: Number Theory and Geometry tasks from MATH [16], one task from AIME 2024 [7], and one task from OlympiadBench [15].

Most reasoning patterns used by QwQ-32B correspond to the most accurate patterns observed for Qwen-2.5-32B, which strongly aligns with our theoretical prediction. We do not report QwQ pattern accuracies because API-based testing does not expose the CoT data needed for the base model to continue generation; consequently, we cannot isolate reasoning-pattern accuracy from overall model accuracy. See the ‘‘Accuracy Analysis’’ procedure in Appendix B.2 for details.

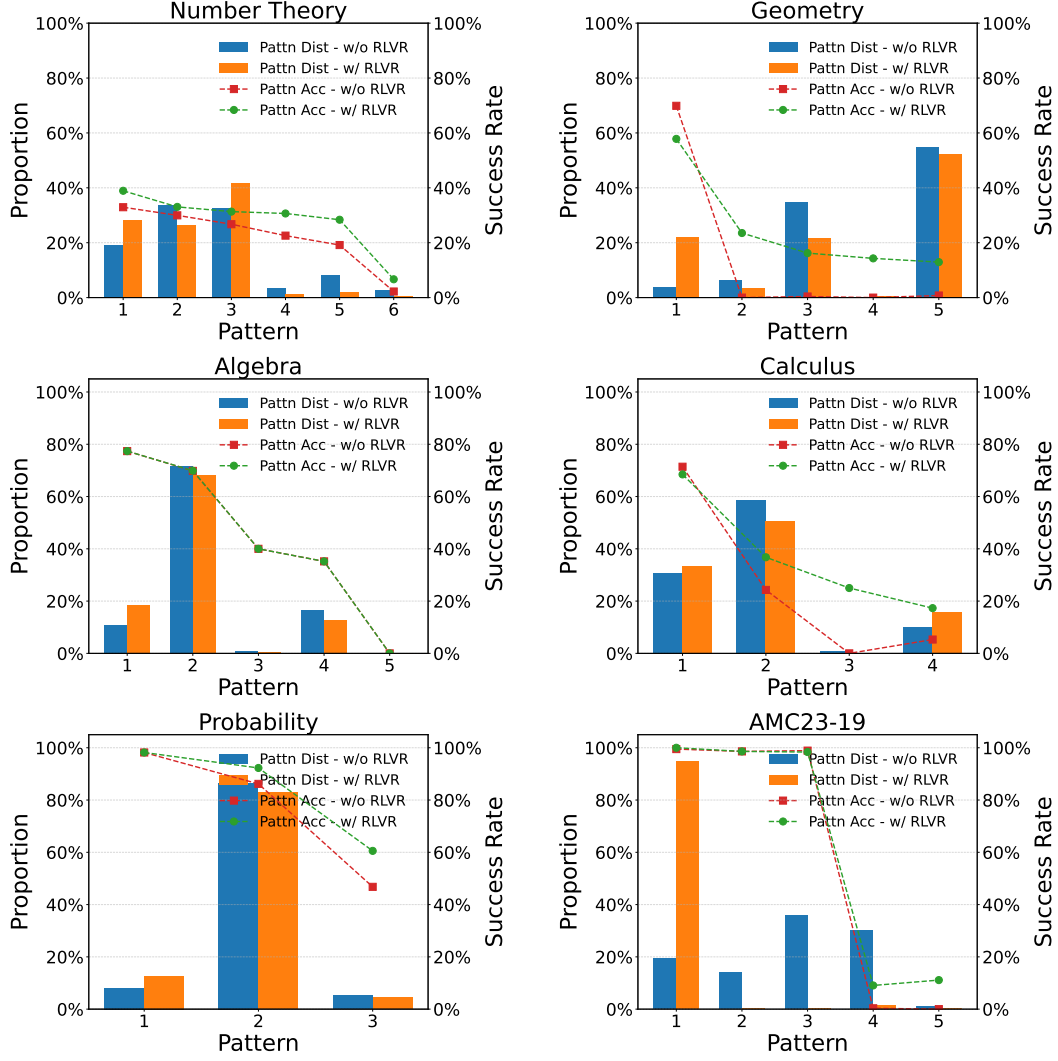
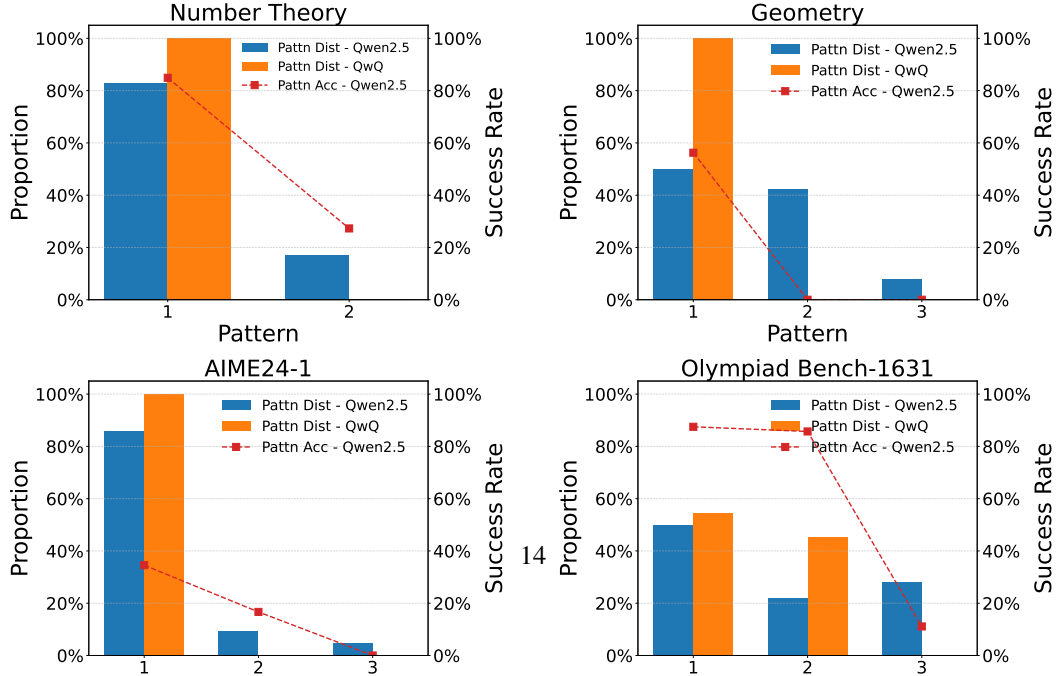


Figure 4: Evaluation results for reasoning patterns and corresponding success rates with and without RLVR enhancement across additional tasks. The results align with our experimental insights and theoretical conclusions. Bars (Pattern Dist) indicate the proportion of each pattern among all patterns, and dots (Pattern Acc) indicate the accuracy or success rate of that pattern.



B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: We use Qwen2.5-3B as the base model and train it on the MATH dataset with both RLVR and RLIF. We adopt the verl [31] framework and keep all parameter settings identical to the MATH example provided by verl. For RLIF, the only change is the reward calculation method. Varying the random seed yields three training curves for each method.

Token-level Analysis: We run this analysis with Qwen2.5-3B and the RL-enhanced variants described above. For GSM8K and MATH, we randomly sample 400 questions from each test set; for AIME24, we use the full dataset. For each question, we first sample the base-model answer, concatenate that answer to the question, and feed the resulting prompt to the RL-enhanced model. We then use the OpenAI library interface to compute each prompt token’s probability and its rank among all possible tokens at that position. Comparing token ranks before and after RL gives the ranking-change proportion reported in the main text.

Reasoning-pattern Level Analysis: We conduct two complementary experiments. First, using Qwen2.5-3B and RL-enhanced Qwen2.5-3B checkpoints, we track pattern-distribution dynamics during training and evaluate every 20 steps. Second, to test whether the conclusions hold for different base models and larger models, we evaluate Qwen-2.5-7B-SimpleRL-Zoo, QwQ-32B, and their corresponding base models. For these external models, we can measure before-and-after changes in output pattern distributions, but not the full training-time dynamics.

B.2 Reasoning Patterns Analysis Pipeline

Model Inference: We randomly sample questions across task types such as Geometry and Algebra and across difficulty levels such as Level 4 and Level 5 from datasets including MATH. For each question, the model samples 1024 answers in the pattern-distribution dynamics experiment. For experiments that only compare distributions before and after RL, the number of sampled answers ranges from 64 to 1024 depending on the task category. These responses form the basis for pattern extraction and classification.

Reasoning Patterns Extraction: We sample a subset of base-model responses and use the GPT-4o [19] API to summarize pattern categories from those samples. The full prompt for reasoning-pattern extraction appears in Appendix D.1. To make extraction deterministic, we set the GPT-4o API temperature to 0.

Responses Classification: We then use GPT-4o to classify all responses into the identified pattern categories, again with temperature set to 0. The full response-classification prompt is provided in Appendix D.2.

Accuracy Analysis: To isolate the effect of pattern distribution on accuracy, we construct new prompts by concatenating the original prompt with partial base-model responses that represent specific reasoning patterns. This fixes the reasoning process, allowing us to attribute accuracy changes to pattern selection rather than to changes inside the reasoning path itself. Concretely, we remove the sentence containing the final answer from the base-model response, append the preceding sentence to the question, and feed the resulting prompt to the RL-enhanced model. We use the accuracy of the continued prediction as the accuracy of the corresponding reasoning pattern.

B.3 Details for Each Task

We now detail the experimental procedures for each task, including task descriptions and examples of reasoning patterns (Table 2). Example task prompts are provided in Appendix D.3.

Number Theory: This task asks models to solve coin-distribution problems across multiple bags. The bags initially contain equal numbers of coins; after additional coins are added and redistributed equally, the total must exceed a specified value while preserving equal distribution. The model

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from } 243) + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Reasoning Patterns Examples for Varying Tasks.

must determine the minimum initial coin count per bag. We derive the task from a level 5 MATH problem (id: test/number_theory/1055) [16] and create 32 evaluation variants by adjusting parameters such as bag count and coin totals. We use one-shot prompting with a simple, unrelated example to enforce "boxed" answer formatting without steering the reasoning strategy. For evaluation, we select 20 questions with 4 responses each for reasoning-pattern extraction, then sample 64 responses per question and categorize all responses by the extracted patterns to compare before-and-after RL distribution shifts.

Geometry: For the geometry task, we give the model the lengths of two sides of a scalene triangle and ask how many integer centimeter values are possible for the third side. The task comes from a level 4 MATH geometry problem (id: test/geometry/1046). The evaluation follows the Number Theory pipeline: we construct a 32-question synthetic dataset, sample 20 questions with 4 base-model responses each from Qwen2.5-7b-Instruct for pattern extraction, and then compare models with and without RLVR enhancement on all evaluation questions using 64 responses per question for pattern analysis.

Algebra: The algebra task is a can-recycling problem. Starting from a fixed number of old cans, the model is told that n old cans can be recycled into one new can and must determine the final number of cans that can be produced. The challenge is that each iterative recycling step can leave extra cans that are not evenly divisible but may later combine to enable additional recycling. The template comes from a level 4 MATH algebra problem (id: test/algebra/2768). For this and the following supplementary tasks, we use the same settings as above, including dataset size and sample counts.

584 **Calculus:** The calculus task presents a single-variable trigonometric equation, specifies the variable
585 domain, and asks the model to determine the number of solutions. Although the task is comparatively
586 simple, it provides a clear probe of basic pre-calculus reasoning. Its prototype is a level 4 MATH
587 pre-calculus problem (id: test/precalculus/1140).

588 **Counting and Probability:** The counting and probability task asks the model to reason about
589 draws from a bin containing white and black balls. Given the probability that two randomly drawn
590 balls contain one ball of each color and the count for one color, the model must compute the minimum
591 possible count for the other color. This task comes from a level 4 MATH counting and probability
592 problem (id: test/counting_and_probability/79).

593 **AMC23 Question:** We use question 19 from AMC23 [1]. For reasoning-pattern extraction, we
594 randomly sample 48 responses; for reasoning-pattern analysis, we sample 1024 responses.

595 B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

596 We conduct the QwQ-32B and Qwen-2.5-32B comparison on four tasks. Because API-based testing
597 limits access, we randomly select one question per task, sample 64 responses from each model, and
598 directly perform reasoning-pattern extraction and classification on those answers. For the MATH
599 tasks, we use question 1055 from Number Theory and question 1046 from Geometry. For the
600 remaining tasks, we use problem 1 from AIME24 and question 1631 from OlympiadBench. Example
601 patterns are shown in Table 2.

602 C Deferred Proofs

603 This appendix gives the complete proofs for our main theoretical results: Proposition 5.2, Theorem 5.3,
604 Theorem 5.4, and Theorem 5.6.

605 C.1 Proof of Proposition 5.2

606 *Proof of Proposition 5.2.* We use the proof strategy of [26]. Recall that the RL optimization objective
607 is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y} | \mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

608 Under our framework from Section 5.1, the policy reasons by first sampling $\mathbf{r}_i \in \mathcal{R}$ according to
609 $\pi_{\theta}(\mathbf{r}_i | \mathbf{q})$ and then producing the final answer $\mathbf{a} \in \mathcal{A}$ according to $\pi_{\theta}(\mathbf{a} | \mathbf{r}_i)$. The objective $\phi_{\text{RL}}(\theta)$
610 can therefore be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{y} | \mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

611 Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$, then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot | \mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot | \mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r} | \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} | \mathbf{r})}{\pi_{\text{ref}}(\mathbf{r} | \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} | \mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot | \mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r} | \mathbf{q})}{\pi_{\text{ref}}(\mathbf{r} | \mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot | \mathbf{q})} \left[\ln \left(\frac{\pi_{\theta}(\mathbf{r} | \mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r} | \mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} | \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

612 The third equality uses Assumption 5.1, and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r} | \mathbf{q})$ is the partition
613 constant ensuring that

$$\pi^*(\mathbf{r} | \mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta}(\mathbf{r} | \mathbf{q}),$$

614 is a valid probability distribution satisfying $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$. The constant Z does not depend on
 615 $\mathbf{r}$.

616 Thus, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned} & \max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ & = \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

617 By the non-negativity and uniqueness properties of KL divergence, the optimal policy for the KL-
 618 constrained reward-maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

619 For RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$. For RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \log |\mathcal{A}| \right)}_c, \end{aligned}$$

620 where the second term is identical for all $\mathbf{r} \in |\mathcal{R}|$. For RLIF, the optimal policy can therefore be
 621 written as

$$\begin{aligned} \pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp \left(\frac{1}{\beta} R'(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i)) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp \left(\frac{1}{\beta} (R'(\mathbf{r}) - c) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp \left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \end{aligned}$$

622 Therefore, $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r}))$. This proves the theorem. $\square$

623 C.2 Dynamics of Tabular Policy

624 Consider the RL objective (Eq 3.1) under the tabular policy (Eq 5.3) with Assumption 5.1. The
 625 gradient dynamics of θ are

$$\begin{aligned} \frac{d}{dt} \theta(t) &= \nabla \phi_{\text{RL}}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})], \end{aligned}$$

626 Here $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to $\mathbf{q}$, and
 627 $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ denotes the KL-regularized reward. Thus,

$$\begin{aligned} \frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i} \right] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})) \right] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \end{aligned}$$

where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t}$$

C.3 Proof of Theorem 5.3

To prove Theorem 5.3, we first show that $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4), then establish a lower bound for $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$ (Eq C.5), and finally derive the time bound stated in Theorem 5.3.

Proof of Theorem 5.3. Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}), \end{aligned}$$

and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) p^*(\mathbf{r}_i) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \quad (\text{C.4}) \end{aligned}$$

Since $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$. For the optimal reasoning pattern $\mathbf{r}^*$, let $\mathbf{r}'$ denote the suboptimal pattern with the second-highest success rate, $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$. Then:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \geq 0. \quad (\text{C.5}) \end{aligned}$$

639 Since $\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq 0$, we have $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})$ for $t \geq 0$. Define $C =$
 640 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^*|\mathbf{q})$:

$$\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.$$

641 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})}}.$$

642 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \epsilon.$$

643 This proves the theorem. $\square$

644 C.4 Proof of Theorem 5.4

645 To prove Theorem 5.4, we first establish an upper bound for $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ (Eq C.6), then analyze the
 646 ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$ (Eq C.7), and finally prove that $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$ for $t > T_0$.

647 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second-best reasoning patterns. For
 648 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned} \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}')) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\ &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2. \end{aligned}$$

649 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

650 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned} \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\ &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0 \end{aligned}$$

651 The last inequality follows from the case-2 condition and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 652 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$ and $t \geq 0$. Since $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

653 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$. We then have:

$$\begin{aligned} \frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}} \end{aligned}$$

654 Then we have:

$$\frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

655 This differential inequality gives:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

656 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

657 Let $C_2 = 1/\Delta$, $C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

658 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

659 Finally, for $t > T_0$ we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

660 This proves the theorem. $\square$

661 **C.5 Proof of Theorem 5.6**

662 *Proof of Theorem 5.6.* First, in the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$, Assumption 5.5 implies
 663 $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of overall accuracy is

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|q) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|q) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, q)) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

664 where $\text{Con}_{\theta(t)} = - \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i|q) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, q))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
 665 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i|q) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, q)) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_j, q)) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, q)) \right). \end{aligned}$$

666 We consider $|\mathcal{R}| \rightarrow +\infty$ with success rates $p^*(\mathbf{r})$ drawn i.i.d. from $U[0, 1]$. By the Central Limit
 667 Theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r}))$ concentrates around its expectation 0.5
 668 with deviation of order $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. Define this positive deviation as $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$, so that
 669 $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$.

670 By the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$, the
 671 empirical distribution of $p^*(\mathbf{r})$ conditioned on the mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$ converges to
 672 the density $f(x) = 1 + 12\delta(x - 0.5)$, $x \in [0, 1]$. We can then approximate the summation over $\mathbf{r}$ by
 673 an integral over $f(x)$, yielding:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3} \delta > 0$$

674 Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this holds with probability 1, proving the
 675 first statement.

To prove the second statement, we analyze the RLIF reward function. For an action space of size
 $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

676 The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing $R_{\text{RLIF}}(\mathbf{r})$ selects the
 677 path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the
 one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

678 Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

679 We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting
 680 distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and
 681 maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables.
 682 Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) = 1 - 6\delta$.
- Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) = 1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta\right) = \frac{1}{2}$$

This concludes the proof of the second statement.

□

D Full Prompts

D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input
Common elements
Pattern 2:
Name and description
Defining characteristics
Example processes from input
Common elements
...
There are reasoning processes:
{extracted_responses}
Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

692

693 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.
{patterns_categories}
Here are the reasoning processes to classify:
{extracted_responses}
Please output your classification in this JSON format:

694

695 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system
You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_endl>
<lim_start>
user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_endl>
<lim_start>assistant
Let me solve this step by step.
<think>

696

Prompt Example for Long Arithmetic

<lim_start>system
You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_endl>
<lim_start>user
Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_endl>
<lim_start>assistant
Let me solve this step by step.
<think>

697

Prompt Example for Number Theory

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

698

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

699

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

700

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

701

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

702

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of ' $8^5 \cdot 5^{10} \cdot 15^5$ '?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

703

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a 9 -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

704

Prompt Example for OlympiadBench-Q1631

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

705